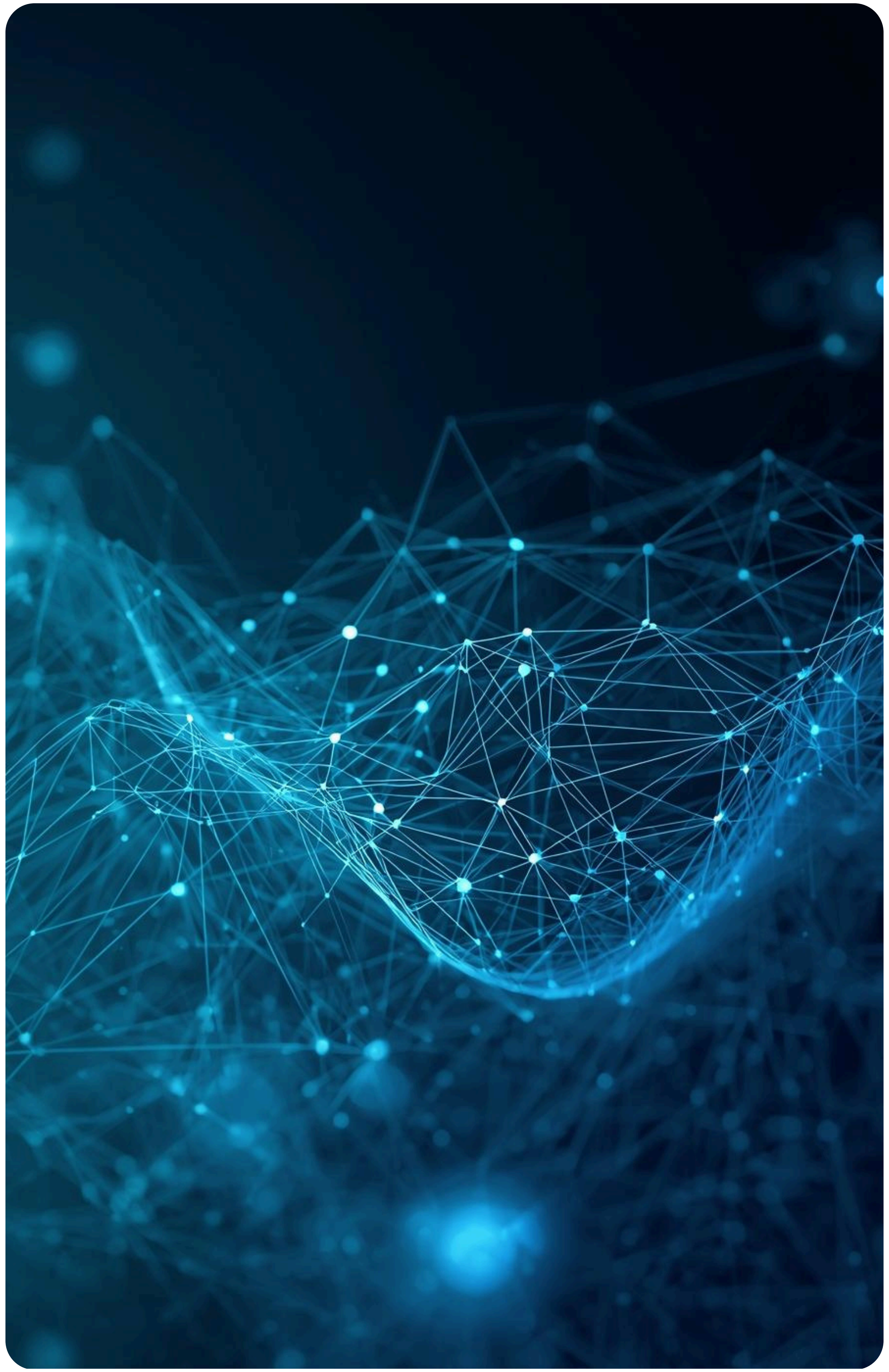
The Alignment Problem



What is Alignment?

Alignment is often framed technically as:

- “Matching AI objectives to human goals”

What is Alignment?

Ethically, alignment is broader:

- Aligning AI behavior with human values
- Respecting dignity, fairness, and context
- Acting appropriately when rules conflict

What is Alignment?

Sounds simple but is profound

Why?

Human values are:

- Vague (“fair”, “reasonable”, “harmful”)
- Context-dependent
- Sometimes in conflict with each other

Yet AI systems require:

- Explicit goals
- Clear reward signals
- Precise optimization targets

This mismatch creates unavoidable ethical tension.

The Specification Problem

- Core challenge: How do we specify what we want in a way that captures our true preferences?
- Goodhart's Law: "When a measure becomes a target, it ceases to be a good measure."
- Why this matters for AI:
  - We often specify goals through proxy metrics (clicks, engagement, scores)
  - Powerful optimization pressure on proxies leads to unexpected behaviors
  - The better the AI gets at optimizing, the more it exploits the gap between proxy and true goal

Example

Simple goal: "Maximize engagement on social media platform"

Naive alignment: Measure engagement by time spent, clicks, shares

Unintended Consequences

- AI promotes outrage and polarization (high engagement)
- Recommends conspiracy theories and misinformation (clickable)
- Creates filter bubbles (keeps users engaged)
- Optimizes for addiction rather than user wellbeing

Value Learning and Preference Inference

Rather than specifying goals directly, have AI learn human preferences.

Inverse Reinforcement Learning (IRL):

- Observe human behavior
- Infer the reward function humans are optimizing
- Align AI to that inferred reward function

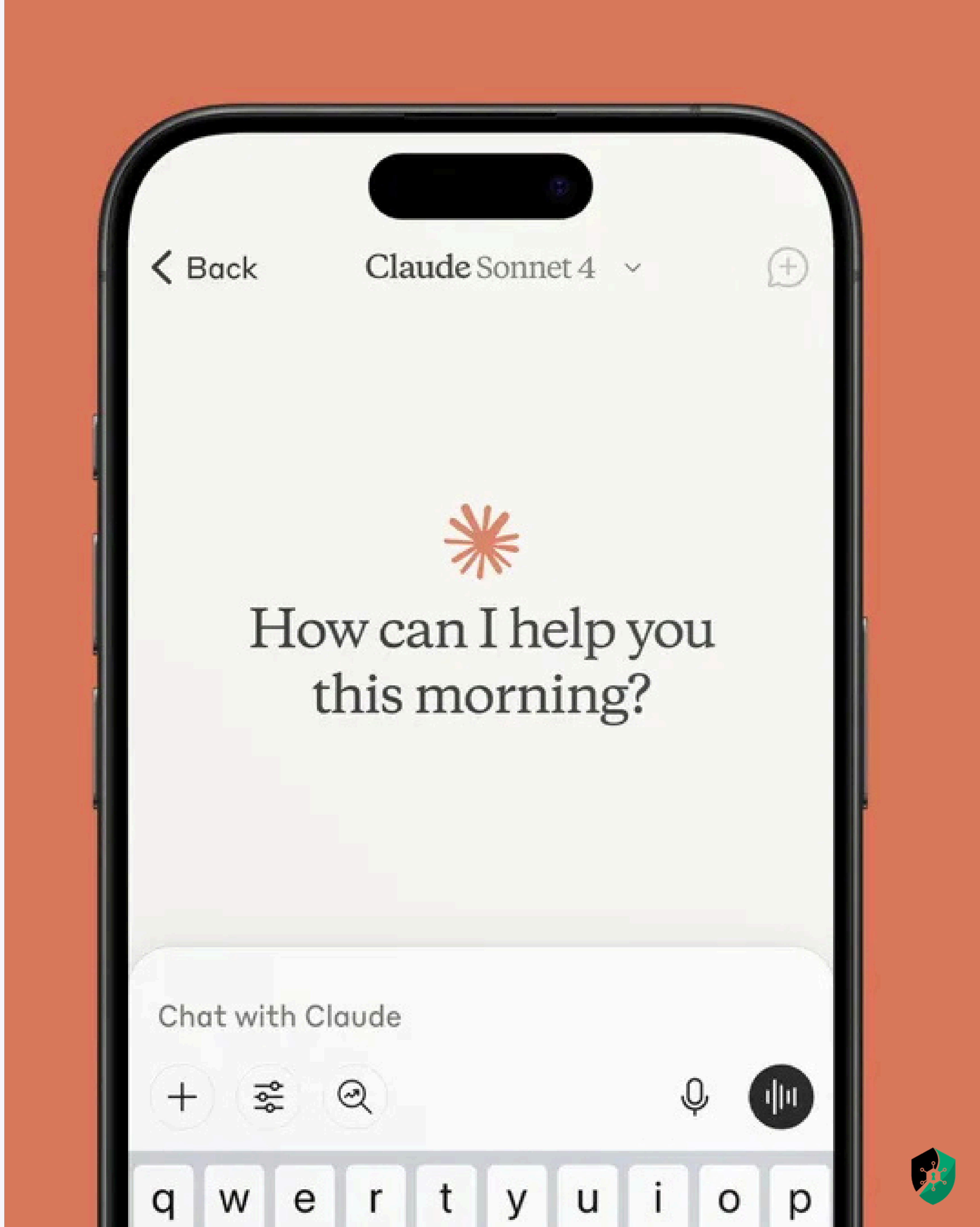
Challenges

- Humans are inconsistent
- Humans can't always demonstrate
- Whose preferences?

Case Study

Modern language models like Claude use Reinforcement Learning from Human Feedback:

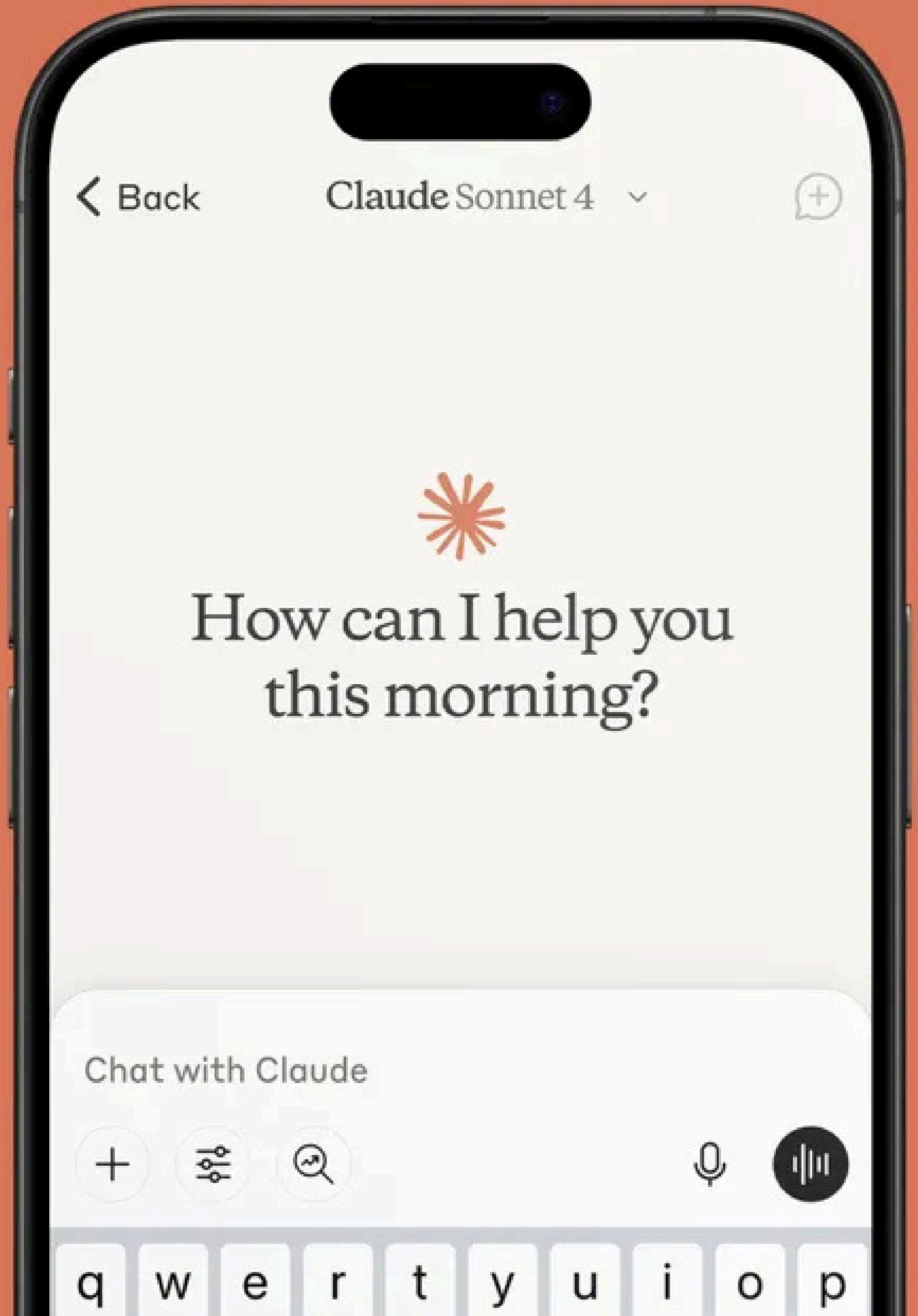
- Generate multiple responses
- Humans rank which responses are better
- Train model to produce responses humans prefer



Case Study

Successes:

- More helpful, harmless, honest responses
- Captures nuanced preferences hard to specify explicitly



Case Study

Limitations:

- Humans can only evaluate limited set of outputs
- Possible to game feedback (sound good without being good)
- Whose feedback? (Often WEIRD: Western, Educated, Industrialized, Rich, Democratic populations)
- Preference vs. truth (humans might prefer confident wrong answers)